

*MA.8.NSO.1.6***Benchmark**

MA.8.NSO.1.6 Solve real-world problems involving operations with numbers expressed in scientific notation.

Benchmark Clarifications:

Clarification 1: Instruction includes recognizing the importance of significant digits when physical measurements are involved.

Clarification 2: Within this benchmark, for addition and subtraction with numbers expressed in scientific notation, exponents are limited to within 2 of each other.

Connecting Benchmarks/Horizontal Alignment

- MA.8.NSO.1.4, MA.8.NSO.1.5

Terms from the K-12 Glossary

- Scientific Notation
- Significant Digits

Vertical Alignment**Previous Benchmarks**

- MA.7.NSO.1.1
- MA.7.NSO.2.1

Next Benchmarks

- MA.912.GR.4.5

Purpose and Instructional Strategies

In grade 7, students developed an understanding of Laws of Exponents with numerical expressions. They focused on generating equivalent numerical expressions with whole-number exponents and rational number bases. In grade 8, students use the knowledge of Laws of Exponents to work with scientific notation. In Geometry, students will solve problems involving density in terms of area and volume which can be represented using scientific notation when the numbers are large. Additionally, students can apply their scientific notation knowledge in science courses.

- Instruction includes opportunities to engage in virtual or physical situations to understand the importance of significant digits.
- Instruction includes student understanding of the following aspects:
 - zeros at the beginning of a number are never significant,
 - zeros at the end of a number are only significant if there is a decimal point and
 - zeros in the middle of a number are always significant.
- Students should develop fluency with and without the use of a calculator when performing operations with numbers expressed in scientific notation.
- For mastery of this benchmark, students are expected to express the product or quotient with the appropriate number of significant digits. In general, the number of significant digits in the result will be the least number of digits in the operands.
 - For example, when multiplying two numbers together, one that has 4 significant digits and the other that has 2 significant digits, then only two significant digits should be retained for the product.

Common Misconceptions or Errors

- Students may incorrectly identify zeros as significant digits.
- Some students may incorrectly apply addition and subtraction across a problem.
 - For example, students may miscalculate $(1.3 \times 10^3) + (3.4 \times 10^5)$ as 4.7×10^8 .
- Some students may incorrectly apply multiplication across a problem.
 - For example, students may miscalculate $(2 \times 10^4)(3 \times 10^5)$ as 6×10^{20} .
- Some students may incorrectly represent their final answer not in scientific notation.
 - For example, students may write $(2.0 \times 10^4)(6.0 \times 10^5)$ as 12.0×10^9 instead of 1.2×10^{10} .

Strategies to Support Tiered Instruction

- Instruction includes making connections of a number written in standard form to the same number written in scientific notation by noticing patterns. Key connections include recognizing the similarities in the first two digits of both numbers and the connections between the place value of the number in standard form and the exponent of the power.
- Teacher provides opportunities for students to utilize appropriate calculators and provides instruction on the various calculator notations for scientific notation.
- Instruction includes rewriting whole numbers in scientific notation when finding products or quotients with scientific notation to demonstrate correct use of operations and laws of exponents.
 - For example, if the student is asked what is five times larger than 2×10^4 , they should be multiplying 5×2 , and not multiply by the exponent.
- Instruction includes making connections to the use of place values when adding and subtracting numbers written in standard form to place values with scientific notation. Teacher should demonstrate how rewriting numbers in scientific notation utilizing the same power of 10 represents numbers with the same place value.
- Instruction includes modeling the correct use of operations and laws of exponents when finding the products and quotients of numbers represented in scientific notation, paying close attention to the solution to ensure it is in scientific notation.
 - For example, when multiplying (3×10^2) and (4×10^4) , students can rearrange the expression as $(3 \times 4)(10^2 \times 10^4)$ to determine 12×10^6 which is equivalent to 1.2×10^7 .
- Teacher provides opportunities for students to check their work by rewriting numbers in standard form and applying any necessary operations before comparing their solution to the solution found with the use of a calculator.

- Instruction includes the use of manipulatives such as Base Ten blocks to make connections to the purpose of utilizing scientific notation.
 - For example, the teacher could pose the question: “What would be the best way for us to represent 2430 using Base Ten Blocks? We could use 2430 individual Base Ten Unit blocks, or we could use 2 Base Ten Cubes, 4 Base Ten Flats and 3 Base Ten Rods. Students can then see that it would be easier to represent 2430 using the Cubes, Flats, and Rods as opposed to the large amount of individual Unit blocks. When students see how it would be easier to use the larger blocks to represent the number, Teachers can explain how it is similar to using scientific notation to write out very large or very small numbers. Instead of writing 287300000000000000, they can write 2.873×10^{18} .
- Teacher provides opportunities for students to complete problems using scientific notation and standard form to check for the reasonableness of their solutions and build on connections between the two.
- Instruction includes the use of a three-read strategy. Students read the problem three different times, each with a different purpose (laminating these questions on a printed card for students to utilize as a resource in and out of the classroom would be helpful).
 - First, read the problem with the purpose of answering the question: What is the problem, context, or story about?
 - Second, read the problem with the purpose of answering the question: What are we trying to find out?
 - Third, read the problem with the purpose of answering the question: What information is important in the problem?

Instructional Tasks

Instructional Task 1 (MTR.6.1)

Measures of population density indicate how crowded a place is by giving the approximate number of people per square unit of area. In 2009, the population of Puerto Rico was approximately 3.98×10^6 people.

Part A. How many significant digits are there in the population of Puerto Rico?

Part B. If the population density was about 1000 people per square mile, what is the approximate area of Puerto Rico in square miles?

Part C. Does the number of significant digits change when finding the population density? Why or why not?

Instructional Items

Instructional Item 1

The Amazon River releases 5.5×10^7 gallons of water into the Atlantic Ocean every second. There are about 3.2×10^9 seconds in a year. How many gallons are released into the ocean in one year? Express your answer with the appropriate number of significant digits.

**The strategies, tasks and items included in the BIG-M are examples and should not be considered comprehensive.*